

Muhammad Wasay

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[Portfolio](#)

Education

Namal University

Bachelor of Science in Computer Science **CGPA: 3.51/4.00**

Pakistan

- Relevant coursework: Data Structures, Algorithms, Software Engineering, Compiler, Computer Vision, Parallel and Distributed Computing, Machine Learning

Technical Skills

Programming Languages: Python, C++, C, Java, CUDA

Tools & Technologies: TensorFlow, Scikit-learn, PyTorch, FastAPI, Docker, Hugging Face, DeepEval, LangChain, Git, GitHub, MySQL, NumPy, Pandas, Matplotlib, Seaborn, Tkinter, AWS, Jenkins, Github Actions, MPI, OpenMP, React.js, Tailwind CSS, Slurm, OpenStack, Kubernetes, NVIDIA CUDA, Prometheus, Grafana

Experience

Phunware (NASDAQ: PHUN)

September 2025 – Present

Platform Architect

Remote – USA

- Architected company-wide LLM evaluation, observability, and reliability infrastructure for production AI systems.
- Implemented DeepEval-based and custom metrics for faithfulness, hallucinations, latency, cost, and task success.
- Developed end-to-end LLM tracing across prompts, tools, agents, and responses for enterprise workloads.

Phunware (NASDAQ: PHUN)

May 2025 – September 2025

AI Team Lead

Remote – USA

- Owned architecture and delivery across legacy systems and new AI platforms, driving platform modernization and engineering velocity.
- Launched **PhunMed**, an AI healthcare assistant for intelligent triage, reminders, and appointment scheduling.
- Integrated **LiveKit** for secure, real-time patient-staff voice communication; scaled to **50k+ users**.

EZ MD Medical

June 2024 - June 2025

Associate Software Engineer

Islamabad

- Designed user-friendly web portals and mobile apps for clients, vendors, and delivery teams, enhancing user experience.
- Led DevOps initiatives, managing CI/CD pipelines and optimizing AWS-based deployments to ensure high availability and performance.

Pakistan Supercomputing

Oct 2022 - May 2025

Lead HPC & Cloud Engineer

Namal University, Mianwali

- Led a high-performance computing project, creating a GPU-based HPC system with 1.2 petaflops capacity for AI and big data research and optimized system performance for large language models (LLaMA-70B), ensuring smooth hardware and software integration for AI research.

DreamBig Semiconductor Inc.

July 2024 - September 2024

Software Engineering Trainee

Karachi

- Contributed to the optimization of network performance by working with RDMA and Smart NICs, including the development of basic kernel and network drivers to enhance system efficiency.
- Collaborated on the development of user-space and kernel-space drivers, memory management and synchronization challenges to improve system stability and performance.

Zindigi Prize

2024 - 2025

Campus Director

Namal University, Mianwali

- Led Zindigi Prize program, driving execution and managing a team of students, entrepreneurs, and mentors for social entrepreneurship initiatives.

Research

Design and Development of RISC-V Based Virtual Cluster using QEMU Simulator | *IEEE Xplore, 2024*

- Designed and developed a high-performance RISC-V-based HPC cluster using the QEMU simulator, comprising a master node and four slave nodes. The research involved utilizing open-source operating systems, distributed programming tools, and benchmarking with a custom compiler toolchain, along with comparison testing against Raspberry Pi and physical RISC-V clusters.

Low-Cost RISC-V-Based High-Performance Computing Cluster for Edge Computing | *ICEPECC 2025*

- Proposed and benchmarked a low-power, scalable RISC-V-based HPC cluster for IoT applications, demonstrating a 63% speedup in matrix operations. Developed a custom distributed logistic regression algorithm for RISC-V, achieving up to 5.6x faster convergence with increased node count.

Projects

Go-To-Market Projects

HPC Cluster Design for Material Design Research at Malakand University | *Slurm, Linux, Python, MPI, Grafana*

- Designed and implemented a 10-node HPC cluster for the Material Science Department at Malakand University, enabling high-throughput while configuring Slurm as the resource management simulations and data analysis for research.

YANA Meals: Insurance-Integrated Food Delivery | *React Native, Node.js, Firebase*

- Architected a 4-application (1 web and 3 mobile apps) suite (Vendor, Rider, Customer, Admin) for insurance-holder meal delivery.
- Secured partnerships with 3 major insurance providers, covering 100K+ potential users.

SmartEmail: AI-Powered Email Agent System | *Python, Google Service API, Gemma 3, FastAPI, React.js, PineCone*

- Developed an AI-powered email agent system for Aricon., automating the processing of emails, organizing them with synthetic and detailed filenames before uploading them to Google Drive and Dropbox and replying to them with a local Gemma 3 model with a vector database.

Industrial Projects

RISC-V HPC Cluster Development | *SiFive SBC, OpenMPI, OpenMP, C++, Python*

- Built an energy-efficient, low-power HPC cluster using RISC-V and Intel NUC, improving large matrix computation speeds by 55%.
- Integrated Wazuh SIEM for security monitoring and federated learning for privacy-preserving decentralized AI model training across institutions.

Multimodal Network Anomaly Detection | *TensorFlow, Scikit-learn, Matplotlib, Pandas*

- Designed a system for anomaly detection in multimodal data streams, including logs and network packets.
- Implemented machine learning algorithms (classifiers, neural networks) to identify unusual patterns in the data.

EdgeGuard: AI-Powered Autonomous Edge Surveillance Framework | *Nvidia Orin, Raspberry Pi, C++, Python, CUDA*

- Developed a real-time AI-powered agentic surveillance framework for edge devices with autonomous event detection and low-latency interactions and queries. ncy video analytics.
- Built a quantization pipeline for AI models to optimize deployment on resource-constrained devices.

Awards

National Tech Expo

First Prize – Heterogeneous ISA Cluster

Developed a heterogeneous ISA cluster for academia and industry; deployed quantized LLMs on a resource-constrained RISC-V cluster, showcasing cutting-edge AI deployment and optimization in limited environments.

Namal Tech Expo

First Place – RISC-V Based Cluster

Built a five-node RISC-V cluster achieving 50% performance gain through efficient parallel task distribution across connected devices.

Namal OpenHouse 2025

Best Final Year Project & Best Poster – EdgeGuard

Recognized for developing **EdgeGuard**, an AI-powered plug and play edge framework featuring real-time event detection, AI as a Service, and quantized AI model deployment on resource-constrained devices.